

IBM FileNet P8 V5.5.8

Detailed Study Material

Deployment, Migration & Certification Preparation

FileNet Deployment Manager (FDM) | Performance & Troubleshooting | IBM C1000-074 Exam Prep

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Section 1 | Learning Objectives

By the end of Day, students will be able to:

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| 1 | Explain the role of FileNet Deployment Manager (FDM) in promoting configurations across environments |
| 2 | Export object store configurations, workflows, document classes, and security settings using FDM |
| 3 | Import FDM export packages into a target environment and validate migrated components |
| 4 | Design a complete environment promotion strategy from Development to Test to Production |
| 5 | Identify and resolve common migration issues including GUID conflicts and missing dependencies |
| 6 | Apply Content Engine performance tuning techniques: JVM heap sizing, connection pooling, caching |
| 7 | Diagnose and resolve workflow performance bottlenecks using Process Engine monitors |
| 8 | Configure and interpret FileNet P8 logging using WAS log levels and CE trace settings |
| 9 | Recall key FileNet P8 architecture, CE administration, workflow, security, and deployment concepts for C1000-074 |
| 10 | Answer practice exam questions across all five C1000-074 exam domains with confidence |

Section 2 | FileNet Deployment Manager (FDM)

2 FileNet Deployment Manager (FDM)

FileNet Deployment Manager (FDM) is an IBM-provided tool for migrating FileNet P8 configurations between environments — typically from Development to Test to Production. FDM exports and imports configuration metadata (not document content) using XML-based deployment packages.

What FDM Does and Does NOT Migrate

FDM DOES migrate: Document classes, property templates, choice lists, entry templates, workflow definitions, storage policies, event subscriptions, security templates, folder structures (empty).

FDM does NOT migrate: Document content (binary files), user/group accounts (from LDAP), application-specific customizations (ICN desktops are migrated separately), WAS configuration.

2.1 Deployment Concepts

2.1.1 Key Terminology

Term	Definition
Deployment Package	An XML-based archive (.zip) created by FDM containing the exported configuration. Portable across FileNet environments.
Source Environment	The environment you are exporting FROM (e.g., Development). Contains the finalized configuration.
Target Environment	The environment you are importing INTO (e.g., Test or Production). Receives the configuration.
Manifest	XML file inside the deployment package listing all exported objects and their dependencies.
Promotion	The act of moving a configuration from a lower environment (Dev) to a higher one (Prod).
Dependency Resolution	FDM automatically identifies and includes dependent objects (e.g., property templates needed by a document class).
Delta Package	An FDM export containing only the changes since the last export. Used for incremental migrations.
Scope	What is included in the export: a single object store, specific classes, specific workflow definitions, or all.

2.1.2 FDM Architecture

FDM is a Java-based tool that communicates with the Content Engine via the Java CEI API. It runs as a standalone command-line tool or GUI and connects to both the source and target environments.

FDM Architecture Components

FDM Client: The FDM application (GUI or command line) running on an admin workstation.

FDM Service: Web service deployed on WAS that handles export/import requests. URL:
http://localhost:9080/fdm/

CE Connection: FDM uses the Java CEI API (MTOM) to connect to both source and target CE servers.

Deployment Package Store: A file system location where export packages are stored. Configurable in FDM.

In our lab: FDM is accessed via the ACCE Admin Console under Tools > Deployment Manager (V5.5.8).

2.2 Exporting Configurations

The export process reads configuration objects from the source Content Engine object store and serializes them into a portable XML deployment package.

2.2.1 Export Scope Options

Export Type	Description	When to Use
Object Store Export	Exports ALL configuration from an object store: all classes, property templates, choice lists, storage policies, event subscriptions, and folder hierarchy.	E3F2FD
Selective Export	Exports specific named objects: individual document classes, specific workflow definitions, or named entry templates. Most common for incremental migrations.	FFFFFF
Incremental Export	Exports only objects modified since the last export date. Requires FDM timestamp tracking. Reduces package size for large environments.	FFFFFF

2.2.2 Export Process via ACCE

1. Log into ACCE on the SOURCE environment: http://localhost:9080/acce (ceadmin / ceadmin)
2. Navigate to: Tools > Deployment Manager > Export
3. Select the Source Object Store: TrainingOS
4. Choose Export Scope: Select 'Entire Object Store' for a full export, or 'Select Items' for selective
5. If selective: Add document classes (e.g., Document, HR_Document), workflow definitions (TrainingWorkflow), and event subscriptions from the available list
6. Configure Package Options: Package Name: TrainingOS_Export_v1.0, Description: Day 10 deployment lab
7. Set Destination Path: C:\FDM_Packages\ (must exist on the FDM server)
8. Click Export. FDM generates the XML package and saves it to the destination path
9. Review the Export Summary report for any warnings or skipped objects

Common Export Warnings and How to Handle Them

'Object references external content' — FDM cannot export the binary content. This is expected. Only metadata is exported.

'Dependency added automatically' — FDM found and included a dependent object (e.g., a property template). Review but generally safe to proceed.

'Object already exists in package' — Duplicate reference. FDM will only include it once. Safe to ignore.

'Workflow definition has active instances' — Export proceeds. Active instances are not affected.

2.3 Importing Configurations

The import process reads a deployment package and creates or updates configuration objects in the target Content Engine object store.

2.3.1 Import Modes

Import Mode	Behavior
Create Only	Creates new objects. Fails if an object with the same symbolic name already exists. Use for first-time imports.
Update Only	Updates existing objects. Fails if an object does not exist. Use for applying changes to already-imported configurations.
Create or Update	Creates if not found, updates if found. Most flexible. Use for repeated promotions.
Replace	Replaces the target object completely with the source version. Overwrites all target customizations.

2.3.2 Import Process via ACCE

1. Log into ACCE on the TARGET environment (can be same server in lab)
2. Navigate to: Tools > Deployment Manager > Import
3. Browse to the package file: C:\FDM_Packages\TrainingOS_Export_v1.0.zip
4. Select Target Object Store: TrainingOS (or destination OS name)
5. Select Import Mode: Create or Update (recommended for lab)
6. Review the Import Preview — FDM shows a list of all objects to be created/updated
7. Check for conflicts: objects with the same name but different GUIDs will be flagged
8. Click Import. FDM processes each object in dependency order
9. Review the Import Summary report — verify all objects show status: Success

2.3.3 Post-Import Validation Checklist

Component	Validation Step
Document Classes	ACCE > Data Design > Document Classes: All exported classes present with correct properties
Property Templates	ACCE > Data Design > Property Templates: All custom properties present
Workflow Definitions	ACCE > Process Configuration > Workflow Definitions: Workflow visible and correct version
Event Subscriptions	ACCE > Events > Subscriptions: All subscriptions present and Enabled status
Storage Policies	ACCE > Storage > Storage Policies: Policy present and assigned to correct classes
Security Templates	ACCE > Security > Templates: Templates migrated with correct ACE structure
Folder Structure	ACCE > Browse: Empty folder hierarchy recreated correctly

2.4 Environment Promotion Strategy

A well-defined promotion strategy ensures consistency and reduces risk when moving configurations from Development to Production.

Recommended 3-Environment Promotion Pipeline

DEVELOPMENT (Dev) --> TEST --> PRODUCTION (Prod)

1. Development: Developers create/modify document classes, workflows, entry templates. FDM export after each sprint.
2. Test: QA team validates migrated configuration. Integration testing with applications. UAT sign-off required.
3. Production: Controlled import during maintenance window. Rollback plan using previous package version.

Key Rules:

- Never make direct changes to Production. Always promote via FDM from Test.
- Tag each package with version and date: TrainingOS_Export_v2.1_20240115.zip
- Store packages in version control (Git) alongside application code.
- Keep a 'rollback' package by exporting Production BEFORE each import.

2.5 Migrated Components – Detail

2.5.1 Object Stores

When migrating object store configuration, FDM captures the logical model but not the physical storage. The following must be re-created manually on the target:

- Storage Area physical paths (e.g., C:\FileNetStore\TrainingOS\FS01 must exist on the target server)
- JDBC Data Source (ECMClientDS) in WAS must be configured pointing to the target DB2 database
- Object Store itself must be pre-created in ACCE before FDM import (FDM does not create the OS)
- Index area and content cache area paths must be re-configured post-import

2.5.2 Workflow Definitions

Workflow definitions exported by FDM include the workflow map XML, step definitions, roster assignments, and launch parameters.

Component	Notes
Workflow Map	Full .pep file content is embedded in the FDM package. Exported from Process Designer.
Roster Assignments	Queue names and roster assignments are migrated. Target PE must have matching roster configured.
Workflow Deadlines	Deadline configurations are included in the workflow map.
Launch Parameters	F_Subject and other launch field defaults are preserved.
Active Instances	NOT migrated. Only the workflow definition template is migrated.

2.5.3 Document Classes and Property Templates

- Document class hierarchy is preserved: base classes and subclasses migrate together
- Property template definitions (name, data type, cardinality, default values) are fully captured
- Choice lists referenced by properties are automatically included as dependencies
- Marking sets and security policy assignments are included
- After import: verify property templates using ACCE > Data Design > Property Templates

2.5.4 Security Configurations

FDM migrates security-related configuration metadata, but does NOT migrate LDAP user/group accounts (those must exist in the target LDAP directory).

- Security Policy definitions are migrated (ACL templates assigned to document classes)
- Marking Sets and their values are included
- Default instance security for classes is migrated
- Object-level ACLs on specific documents/folders are NOT migrated (they stay with the content)
- Post-migration: Re-assign LDAP groups to roles because group GUIDs differ between environments

Section 3 | Performance & Troubleshooting

3 Performance & Troubleshooting

3.1 Content Engine Performance Tuning

Content Engine performance depends on four main areas: JVM memory allocation, database query performance, connection pool settings, and content cache effectiveness.

3.1.1 JVM Heap Sizing for WAS/CE

WAS JVM Heap Configuration for FileNet P8 V5.5.8

Location: WAS Admin Console > Servers > Server1 > Process Definition > Java Virtual Machine

Recommended Settings (Training Lab - 8GB RAM server):

Initial Heap Size (-Xms): 512 MB

Maximum Heap Size (-Xmx): 2048 MB

Production Recommendations (16GB+ server):

Initial Heap Size (-Xms): 1024 MB

Maximum Heap Size (-Xmx): 4096 MB

Additional JVM Arguments for FileNet:

-Xgcpolicy:gencon (generational garbage collection - best for CE workloads)

-Xmn512m (nursery space for young objects)

-XX:+DisableExplicitGC (prevent application-triggered GC calls)

3.1.2 Database Connection Pool (DB2)

Parameter	Recommended Value & Notes
Min Connections	Minimum DB connections kept open. Set to 10 for development, 25 for production.
Max Connections	Maximum DB connections allowed. Set to 50 for development, 150 for production.
Connection Timeout	Time to wait for a free connection before throwing error. Default: 180 seconds. Reduce to 30s in production.
Reap Time	Interval between connection pool maintenance runs. Default: 180 seconds.
Unused Timeout	Time before idle connections above minimum are closed. Default: 1800 seconds.
Location	WAS Admin Console > Resources > JDBC > Data Sources > ceadmin_DS > Connection Pool

3.1.3 Content Cache Optimization

- Cache Size: Set to 2-4x the average daily retrieval volume. Monitor via ACCE > Storage > Cache Statistics
- Cache Location: Place on local SSD, never on network storage. Path should NOT be on the same disk as the OS
- Eviction Policy: LRU (Least Recently Used) — built in, not configurable. Ensure adequate sizing to avoid thrashing
- Cache Area Online: Verify ACCE > Storage Areas > Cache Area shows status Online
- Multiple Cache Areas: In clustered WAS, configure a separate cache area per server node

3.1.4 CE SQL Query Optimization

Slow CE SQL queries are one of the most common performance issues. Key optimization techniques:

Technique	Description
Use PropertyFilter	Only request properties you need. Fetching all properties (null filter) is expensive for large result sets.
Limit Result Sets	Always set a maxItems value in fetchObjects(). Never fetch unlimited results in production.
Index Custom Properties	For custom properties used in WHERE clauses, enable indexing on the property template.
Avoid LIKE with leading %	SELECT * FROM Document WHERE title LIKE '%keyword' cannot use indexes. Rewrite as full-text search.
Use INFOLDER with path	Filter results to a folder subtree to reduce the search scope on large object stores.
Batch Updates	Update multiple documents in a single CE transaction rather than one-at-a-time saves.

3.2 Workflow Performance

Process Engine performance depends on queue depth management, workflow complexity, and the PE server's ability to process work items in parallel.

3.2.1 Key PE Performance Metrics

Metric	Target & Notes
Queue Depth	Number of work items waiting in a queue. Target: < 100 for real-time queues. Monitor via ACCE > Workflow Monitor.
Step Processing Time	Average time to complete one workflow step. Monitor via PE metrics. Target varies by business process.
Dispatch Thread Pool	Number of threads PE uses to dispatch work items. Default: 10. Increase to 20-40 for high-volume queues.
Roster Capacity	Max concurrent workflow instances. Ensure the roster has sufficient capacity for peak load.
Database Wait Time	Time PE spends waiting for DB2 responses. High DB wait = DB2 performance problem, not PE.
Workflow Map Complexity	Deep workflow maps with many conditional branches increase processing time.

3.2.2 PE Performance Tuning Steps

1. Check Queue Depth: ACCE > Process Configuration > Work Queues > [QueueName] > Properties > Current Depth
2. Increase dispatch threads: PE Admin Console > Server > Thread Pool > Dispatch Threads (default 10, try 20)
3. Review workflow map for unnecessary wait steps or large attachment transfers in workflow data fields
4. Use asynchronous event subscriptions (Post-processing) instead of synchronous (Pre-processing) for workflow launch
5. Partition high-volume workflows across multiple isolated regions to distribute load

3.3 Logging and Monitoring

Effective logging is critical for diagnosing FileNet P8 issues in production. Log levels must be balanced — too verbose impacts performance, too quiet misses errors.

3.3.1 Key Log Files

Log File	Contents & Location
WAS SystemOut.log	Primary log. CE startup, authentication, JDBC, API errors. Location: <WAS_HOME>\profiles\FileNetServer\logs\server1\
WAS SystemErr.log	Java exceptions and stack traces. Review when SystemOut.log shows truncated errors.
CE Trace Log	Detailed CE-specific trace. Enabled via ACCE > Admin > Logging. Very verbose — enable only for debugging.
PE Server Log	Process Engine events, workflow routing, step execution. Location: <PE_HOME>\log\
FDM Log	FDM export/import operation log. Location: <FDM_HOME>\logs\fdm.log
DB2 Diagnostic Log	DB2 database errors and JDBC timeout details. Location: <DB2_HOME>\DB2\logs\db2diag.log

3.3.2 Configuring Log Levels in WAS

Log levels control verbosity. For production use INFO. For debugging use FINE or FINEST on specific packages.

```
# WAS Admin Console > Troubleshooting > Logs and Trace > server1 > Diagnostic Trace
# Click Change Log Detail Levels and add:

# For CE API debugging:
com.filenet.api.*=FINE

# For JDBC/DB2 connection debugging:
com.ibm.ws.rsadapter.*=FINE

# For security/authentication debugging:
com.ibm.ws.security.*=FINE

# Reset to default (production):
*=INFO

# CE-specific trace via ACCE:
# ACCE > P8Domain > Administration > Logging > Add Trace Logger
# Logger: com.filenet.engine    Level: WARNING (production) / DEBUG
(troubleshooting)
```

3.3.3 Common Issues and Diagnostic Steps

Symptom	Likely Cause	Resolution
Document retrieval slow	Cache miss rate > 20%, or storage area on slow network disk	Check ACCE cache stats. Move storage area to local SSD. Increase cache size.
CE API timeout	DB2 connection pool exhausted or network latency	Increase DB connection pool max. Check DB2 diagnostic log for lock waits.
Workflow not launching from event subscription	PE service stopped or PE_ConnectionPoint offline	Check PE service status. Verify connection point in ACCE. Review SystemOut.log.
FDM import fails with GUID conflict	Target OS has an object with same name but different GUID	Use 'Create or Update' import mode. Review manifest to identify conflicting objects.
LDAP authentication fails after migration	Bind DN credentials different in target environment	Re-configure VMM LDAP in WAS Admin Console for the target server's LDAP.
CE log shows OutOfMemoryError	WAS JVM heap too small for workload	Increase -Xmx to 4096m. Enable garbage collection logging to identify leak.
Search queries return no results	Event subscriptions or index queue backed up	Check CE index area in ACCE. Verify full-text index service is running.

Section 4 | Certification Preparation | IBM C1000-074

4 Certification Preparation – IBM C1000-074

The IBM Certified Administrator – FileNet Content Manager V5.5 exam (C1000-074) tests knowledge across five domains. This section provides a comprehensive review with sample questions and answers mapped to each domain.

C1000-074 Exam Overview

Exam Code: C1000-074

Full Title: IBM Certified Administrator - FileNet Content Manager V5.5

Questions: 63 multiple-choice questions

Passing Score: 47 correct (approximately 75%)

Duration: 90 minutes

Domains: Architecture (20%), CE Administration (26%), Workflow (20%), Security (18%), Deployment (16%)

4.1 Domain 1: FileNet Architecture (20%)

Know the components, their roles, and how they interconnect in a FileNet P8 V5.5 deployment.

Component	Role
Content Platform Engine (CPE)	Core server managing all object stores, documents, events. Deployed as EAR on WAS. Database: DB2 or Oracle.
Process Engine (PE)	Manages workflow routing, work queues, rosters. Communicates with CPE via RMI/IIOP on port 32768.
IBM Content Navigator (ICN)	Web UI for end users. Deployed separately on WAS. Connects to CE via MTOM or IIOP.
ACCE	Administration Console for Content Engine. Browser-based admin tool for CE configuration.
WAS (WebSphere App Server)	J2EE container hosting CPE, ICN, FDM, and CMIS services.
DB2 / Oracle	Relational database storing CE metadata. Three databases: GCDDDB (domain), OSDB (object store), PEDB (PE).
LDAP / VMM	User authentication via WebSphere Virtual Member Manager. Connects to LDAP for user/group lookup.
FDM	FileNet Deployment Manager. Migrates configuration metadata between environments.

Q: What protocol does the Process Engine use to communicate with the Content Engine?

A: RMI/IIOP on port 32768

Options: A) HTTP/80 B) RMI/IIOP/32768 C) MTOM/9080 D) JDBC/50000

Q: Which database stores the Global Configuration Data (GCD) for a FileNet P8 domain?

A: GCDDDB — the GCD database stores domain-level configuration including object store definitions and security settings.

Options: A) OSDB B) GCDDDB C) PEDB D) CEDB

4.2 Domain 2: Content Engine Administration (26%)

This is the highest-weighted domain. Focus on object store creation, document classes, property templates, and storage management.

Topic	Key Facts to Remember
Object Store Creation	Created via ACCE wizard. Requires OSDB database, storage area, and index area configured before creation.
Document Classes	Inherit from Document base class. Custom properties added via property templates.
Property Templates	Reusable property definitions. Types: String, Integer, DateTime, Boolean, Object, Binary.
Choice Lists	Fixed set of allowed values for a property. Created in ACCE > Data Design > Choice Lists.
Storage Areas	File System (most common), FCD (WORM), Advanced, Database. Set in ACCE > Storage Areas.
Storage Policies	Route documents to storage areas. Assigned per class or per object store.
Content Cache	Accelerates retrieval. Monitor hit ratio via ACCE > Storage > Cache Statistics. Target: > 80%.
Marking Sets	Classify document sensitivity. Used with security policies for access control.

Q: What is the minimum number of storage areas required to create a FileNet P8 object store?

A: One storage area (default storage area) must be assigned to the object store during creation.

Options: A) Zero B) One C) Two D) Three

Q: Which property template data type should be used to store a document's approval date?

A: DateTime — used for date and time values. Integer is for numbers, String for text.

Options: A) String B) Integer C) DateTime D) Boolean

4.3 Domain 3: Workflow Management (20%)

Term	Definition
Process Engine (PE)	Manages workflow lifecycle. Communicates with CE via connection point. Port: 32768.
Connection Point	Named link between CE and PE. Created in ACCE > P8Domain > Process Engine > Connection Points.
Isolated Region	Groups workflows for performance isolation. Region 1 is the default.
Workflow Definition	The workflow template (map) designed in IBM Process Designer. Exported as .pep file.
Workflow Instance	A running execution of a workflow definition. Created on launch. Has unique work object number.
Queue (Roster)	Container for work items waiting for human action. Every step that involves user action needs a queue.
Work Item	A single task in a workflow instance. Resides in a queue. A user claims and steps it.
Event Subscription (Process Router)	Automatically launches a workflow when a FileNet document event fires (e.g., OnCheckin).

Q: What is the purpose of an Isolated Region in the Process Engine?

A: An Isolated Region is a logical group of workflows on a PE server that provides performance isolation. Workflows in one region do not share resources with workflows in another region.

Options: A) To isolate test workflows from production B) To group workflows for performance isolation on a PE server C) To prevent users from accessing certain queues D) To partition object stores

4.4 Domain 4: Security Configuration (18%)

Concept	Key Facts
ACL (Access Control List)	Every securable object has an ACL. Made up of ACEs (Allow/Deny per principal).
ACE (Access Control Entry)	One entry in an ACL. Specifies: Grantee (user/group), Access Type (Allow/Deny), Permissions (bitmask).
Deny precedence	A Deny ACE always overrides an Allow ACE for the same principal. No exceptions.
Special Identities	#AUTHENTICATED-USERS, #CREATOR-OWNER, #EVERYONE. Cannot be used for Allow/Deny on sensitive objects.
Security Inheritance	Child objects inherit ACEs from parent (folder or class). Inheritance can be broken per object.
LDAP / VMM	WAS VMM federates LDAP. CE resolves group memberships at runtime via VMM API calls.
LTPA Token	WAS-issued encrypted cookie for SSO. Validated by CE on every API call. Default timeout: 30 min.

Marking Sets	Sensitivity labels (e.g., Confidential, Public). Combined with security policies to control access.
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Q: A user belongs to two groups: Group_Allow has Allow:Read on a document, and Group_Deny has Deny:Read. What is the effective access?

A: Deny takes precedence. The user cannot read the document because a Deny ACE overrides any Allow ACE, regardless of group membership order.

Options: A) Allow — Allow takes precedence B) Deny — Deny takes precedence C) Depends on order D) No access — both cancel out

4.5 Domain 5: Deployment Tools (16%)

Topic	Key Facts
FDM Package	ZIP archive of XML files. Contains manifest, object definitions, dependency list.
Import Modes	Create Only, Update Only, Create or Update, Replace. 'Create or Update' is safest for repeated migrations.
What FDM migrates	Document classes, property templates, workflows, event subscriptions, storage policies, security templates.
What FDM does NOT migrate	Document content (binary), LDAP accounts, WAS configuration, object store physical storage paths.
Dependency resolution	FDM automatically includes dependent objects in the export package.
Rollback strategy	Export target environment BEFORE import. Store as rollback package. Re-import if issues arise.
Delta migration	Export only objects modified after a given date using FDM incremental export.
Post-import steps	Verify storage area paths, reassign LDAP groups, re-configure WAS LTPA keys, test application connectivity.

Q: Which of the following is NOT migrated by FileNet Deployment Manager (FDM)?

A: Document content (binary files stored in storage areas) is NOT migrated. FDM only migrates configuration metadata (classes, properties, workflows, etc.).

Options: A) Document classes B) Document content (binary files) C) Workflow definitions D) Storage policies

4.6 Quick Reference – Exam Facts

Item	Value	Notes
CE DB port	DB2 default: 50000, Oracle: 1521	Always verify in JDBC data source
PE port	32768 (RMI/IIOP)	Used for CE-to-PE communication
WAS HTTP	9080	CE MTOM, CMIS, ICN, ACCE
WAS HTTPS	9443	Secure access, ACCE HTTPS
WAS Admin HTTP	9060 / HTTPS 9043	Admin console
LDAP port	389 (clear) / 636 (SSL)	VMM LDAP repository
ICN Config URL	http://host:9080/navigator/config	ICN configuration tool
FDM service URL	http://host:9080/fdm/	FDM web service endpoint
ACCE URL	http://host:9080/acce	Admin console for CE
CMIS Browser	http://host:9080/fncmis/browser	REST/CMIS API endpoint
PE REST	http://host:9080/pe-rest/v1/	Workflow REST API
CE MTOM	http://host:9080/wsi/FNCEWS40MTOM/	Java/MTOM API endpoint

Section 5 | 30 Practice Exam Questions | IBM C1000-074

5 30 Practice Exam Questions

The following questions are representative of the IBM C1000-074 exam style and difficulty. Each question includes the correct answer and explanation.

Architecture (Questions 1-6)

Q: 1. Which component of FileNet P8 is responsible for storing and retrieving document binary content?

A: The Content Platform Engine (CPE) manages document storage. The actual binary content is stored in Storage Areas (file system, FCD, etc.) managed by CPE.

Options: A) Process Engine B) Content Platform Engine (CPE) C) IBM Content Navigator D) WAS Application Server

Q: 2. In a standard FileNet P8 V5.5 deployment, how many separate databases are typically required?

A: Three databases: GCDDb (Global Configuration Data for the domain), OSDB (Object Store metadata), and PEDb (Process Engine data).

Options: A) One B) Two C) Three D) Four

Q: 3. What is the role of WebSphere Virtual Member Manager (VMM) in FileNet P8?

A: VMM provides a federated identity layer in WAS that connects FileNet to one or more LDAP directories. CE queries VMM at runtime to resolve group memberships for ACL evaluation.

Options: A) Store document content B) Provide federated user/group identity from LDAP C) Manage workflow queues D) Host the ACCE admin console

Q: 4. IBM Content Navigator (ICN) connects to Content Engine using which protocol by default?

A: ICN connects to CE using the MTOM web services protocol over HTTP (port 9080). The CE URL is configured in ICN's repository connection settings.

Options: A) IIOP/2809 B) MTOM/9080 C) JDBC/50000 D) REST/443

Q: 5. Which FileNet P8 component handles LTPA token validation on API calls?

A: The Content Engine (CPE) validates the LTPA token on every API call by decrypting it with the shared LTPA key configured in WAS.

Options: A) Process Engine B) LDAP server C) Content Platform Engine (CPE) D) IBM Content Navigator

Q: 6. What is the maximum number of object stores supported per FileNet P8 domain in V5.5?

A: FileNet P8 V5.5 supports up to 256 object stores per domain. Practical production deployments typically use 1-10 object stores per domain for manageability.

Options: A) 10 B) 64 C) 128 D) 256

CE Administration (Questions 7-14)

Q: 7. When creating a new document class in ACCE, what base class must all document classes ultimately derive from?

A: All document classes in FileNet P8 must derive from the base class 'Document' (system class). The inheritance chain can have multiple levels.

Options: A) Object B) CmAbstractPersistable C) Document D) IndependentObject

Q: 8. A property template is set to cardinality 'LIST'. What does this mean?

A: A cardinality of LIST means the property can hold multiple values (an ordered list of values of the defined data type). This is in contrast to SINGLE (one value) or ENUM (fixed set).

Options: A) The property value must be from a choice list B) The property stores a single value C) The property can store multiple values in an ordered list D) The property is read-only

Q: 9. What is the Affinity Cost of a storage area and how is it used by storage policies?

A: Affinity Cost is a numeric value (0-100) assigned to a storage area. Storage policies use affinity cost to prioritize storage areas: lower cost = higher preference for writing new content.

Options: A) The maximum file size stored in that area B) A priority value where lower cost = higher preference C) The cost of disk storage per GB D) The number of concurrent connections allowed

Q: 10. Which command in DB2 verifies that the GCDDb database was created successfully?

A: 'db2 LIST DB DIRECTORY' lists all databases on the DB2 instance, confirming GCDDb (and OSDB, PEDb) exist.

Options: A) db2 SHOW DATABASES B) db2 LIST DB DIRECTORY C) db2 CHECK DATABASE GCDDb D) db2 DESCRIBE DATABASE

Q: 11. What is the purpose of an Entry Template in FileNet P8?

A: An Entry Template is a pre-configured document add form that sets default property values, restricts which properties users see, and controls which document class is used. Configured in ACCE and used in ICN.

Options: A) Define the physical storage location B) Pre-configure document add forms with default values and restricted fields C) Control workflow routing rules D) Map users to security roles

Q: 12. A Fixed Content Device (FCD) storage area is required for which regulatory compliance scenario?

A: FCD storage provides WORM (Write Once Read Many) compliance required for SEC Rule 17a-4 (financial records retention), HIPAA (healthcare), and SOX (financial reporting).

Options: A) High-speed document retrieval B) WORM compliance for SEC 17a-4/HIPAA/SOX C) Encrypting document content D) Replicating content to DR sites

Q: 13. What happens when a document class is deleted from ACCE while documents of that class still exist in the object store?

A: The operation fails. You cannot delete a document class that has existing instances. All documents of that class must be reclassified or deleted first.

Options: A) All documents are automatically deleted B) Documents are reclassified to the Document base class C) The operation fails — existing instances prevent deletion D) The class is marked inactive

Q: 14. Which FileNet P8 V5.5 URL provides access to the Administration Console for Content Engine (ACCE)?

A: ACCE is accessible at `http://<server>:9080/acce`. It uses the same WAS HTTP port as CE. Credentials are the CE admin account (ceadmin).

Options: A) `http://host:9080/navigator` B) `http://host:9080/acce` C) `http://host:9080/filenet/Engine` D) `http://host:9043/acce`

Workflow Management (Questions 15-20)**Q: 15. In Process Designer, what is the Launch Step in a workflow definition?**

A: The Launch Step is the entry point of a workflow map. It defines the initial data fields required to start the workflow and transitions to the first user or system step.

Options: A) The step where documents are attached B) The entry point that defines launch parameters and transitions to the first step C) A step that automatically launches sub-workflows D) The final approval step

Q: 16. What is the difference between a Roster and a Queue in Process Engine?

A: A Roster is a configuration container for queues on a PE server. A Queue (also called a work queue) is where work items are held awaiting user action. A roster contains one or more queues.

Options: A) They are the same thing B) A Roster contains queues; a Queue holds individual work items C) A Queue contains rosters; a Roster holds work items D) Rosters are for automatic steps; Queues for manual steps

Q: 17. How can an event subscription be configured to automatically launch a workflow when a document is checked in?

A: Create an Event Subscription with Event = OnCheckin, and set the Event Action type to Process Router. The Process Router points to the workflow definition and maps document properties to workflow data fields.

Options: A) Use a Scheduled Event with a cron expression B) Create an Event Subscription with OnCheckin event and a Process Router Event Action C) Configure the workflow to poll for new document check-ins D) Use a CE API scheduled job

Q: 18. What is the work object number in Process Engine?

A: The work object number is the unique identifier assigned to each workflow instance when it is launched. It is used to retrieve, step, and track a specific workflow instance via API or UI.

Options: A) The number of work items in a queue B) The unique identifier of a workflow instance C) The PE server port number D) The workflow step sequence number

Q: 19. A workflow is stuck and no work items appear in the queue. What is the first diagnostic step?

A: Check the Process Engine service status (running/stopped) in Windows Services or via the PE Admin Console. Also verify the PE_ConnectionPoint is Online in ACCE.

Options: A) Delete and re-create the workflow B) Restart the WAS server immediately C) Check PE service status and verify PE_ConnectionPoint is Online in ACCE D) Increase queue capacity

Q: 20. Which PE configuration setting controls how many workflow steps can be processed simultaneously?

A: The Dispatch Thread Pool size controls parallelism. The default is 10 threads. Increasing this value allows more workflow steps to be processed in parallel on a single PE server.

Options: A) Queue Depth Limit B) Isolated Region Count C) Dispatch Thread Pool Size D) Roster Capacity

Security (Questions 21-25)

Q: 21. What is the effect of setting 'Restrict Permission Inheritance' on a document's ACL?

A: It breaks the inheritance chain for that document. The document's ACL becomes independent of its parent folder. Changes to the parent folder ACL will no longer propagate to this document.

Options: A) The document can only be accessed by administrators B) The document no longer inherits ACEs from its parent folder C) The document's ACL is deleted D) All users are denied access

Q: 22. Which FileNet special identity represents the user who created a specific document?

A: #CREATOR-OWNER is a special identity that resolves to the user who created a particular document instance. It is evaluated per-object at runtime.

Options: A) #AUTHENTICATED-USERS B) #EVERYONE C) #CREATOR-OWNER D) #DOCUMENT-OWNER

Q: 23. An administrator wants to prevent any user from deleting documents in a folder, even if they have Full Control from a parent ACL. What is the correct approach?

A: Add a Deny ACE for Delete permission on the folder for #AUTHENTICATED-USERS. Deny ACEs always override Allow ACEs, so even users with Full Control from a parent will be denied Delete.

Options: A) Remove the folder from all ACLs B) Add a Deny ACE for Delete on the folder for #AUTHENTICATED-USERS C) Set folder accessibility to Read-Only in ACCE D) Disable the Delete permission in WAS security roles

Q: 24. What is the LTPA key timeout default in WebSphere Application Server and where is it configured?

A: The default LTPA token timeout is 30 minutes. It is configured in WAS Admin Console under Security > Global Security > LTPA > Token Timeout.

Options: A) 15 min, in ACCE > Security B) 30 min, in WAS Admin Console > Security > Global Security > LTPA C) 60 min, in CE properties file D) 120 min, in WAS server.xml

Q: 25. In VMM LDAP configuration, what is the purpose of the Bind DN?

A: The Bind DN is the service account (Distinguished Name) that WAS VMM uses to authenticate to the LDAP server in order to perform user and group search operations. It must have read access to the user and group OUs in LDAP.

Options: A) The LDAP administrator account for making changes B) The service account VMM uses to connect to LDAP and search for users/groups C) The base DN under which all users are stored D) The DN of the FileNet admin user in LDAP

Deployment Tools (Questions 26-30)

Q: 26. You exported a deployment package from Development. When you import it to Production using 'Create Only' mode, an error occurs: 'Object already exists.' What should you do?

A: Switch the import mode to 'Create or Update'. This will create any new objects and update any existing ones, handling both scenarios in a single import pass.

Options: A) Delete the conflicting objects from Production first B) Switch import mode to Create or Update C) Use Replace mode to overwrite everything D) Re-export from Development with a different package name

Q: 27. What must be done BEFORE running an FDM import into a new (empty) target object store?

A: The target object store must already exist in ACCE (FDM does not create object stores). Storage areas, index areas, and JDBC data sources must also be configured on the target server.

Options: A) Nothing — FDM creates the object store automatically B) The target object store must already exist; storage areas and JDBC data sources must be configured C) Only the target database needs to exist D) Install the same version of FDM as the source

Q: 28. Which FDM export type includes only configuration objects modified after a specific date?

A: Delta (Incremental) Export. This captures only objects with a modification timestamp after the specified cutoff date, reducing package size for environments with large object stores.

Options: A) Full Export B) Selective Export C) Delta (Incremental) Export D) Differential Export

Q: 29. After an FDM migration, LDAP group assignments on document classes are missing in the target environment. What is the most likely cause?

A: FDM migrates security templates and ACL structure, but LDAP group accounts are not migrated. The target LDAP directory may have the groups with different GUIDs than the source, causing the ACL references to be unresolvable.

Options: A) FDM never migrates security configurations B) LDAP groups have different GUIDs in the target environment C) The target object store is in read-only mode D) The security templates were excluded from the export scope

Q: 30. What is the recommended approach to handle a failed FDM import that caused partial configuration in the target?

A: Import the rollback package (a full export of the target taken before the failed import). This restores the target to its pre-import state. Then diagnose the original package issue and retry.

Options: A) Manually delete the partial objects and retry B) Re-import the same package with Replace mode C) Import the rollback package to restore the target's pre-import state D) Restore the target database from backup